Task 4 - Sandstorm Traverse (Anti-Stall)

Wokwi link: https://wokwi.com/projects/479661110505954305

What it does

An ESP32 drives a left and right motor (shown as two green LEDs), reads the motor current (a slide potentiometer: 0 V = 0 A, 3.3 V = 30 A), and frees the rover if a wheel jams.

How to use

Press play, slide the potentiometer, and watch the serial monitor. Type 0.8 0.0 (speed, turn) or stop to give commands.

How it works

- **No delay().** The code uses millis() timers, so nothing ever blocks.
- **Fast loop (every 10 ms = 100 Hz):** reads current, works out wheel speeds, and checks for a stall.
- **Slow loop (every 50 ms = 20 Hz):** reads serial commands and prints telemetry.
- **Steering:** Left = v - w*width/2, Right = v + w*width/2.

Stall detection and battery protection

Current	Result
below 10 A	normal driving
above 15 A	WARNING shown in telemetry
above 20 A for more than 350 ms	STALL

When a stall is detected:

1. Power drops to 20% (red LED on), so the drivers and battery stop being stressed.
2. Motors reverse for 200 ms to back off the rock.
3. STALL ACTIVE is printed.

The rover stays at 20% power until the current falls under 10 A, then returns to full power. If it is still stuck, the same steps repeat.

The 350 ms wait stops short current spikes (like motor start-up) from counting as a stall.